STUDYNEST | CLASS 11 PHYSICS

QUESTION PAPER 1

Chapter 6: System of Particles and Rotational Motion

Maximum Marks: 40 | Suggested time: 60 minutes

Instructions: Answer all questions. Show calculation steps and units. Original practice paper, not an official examination.

Section A 1-mark questions - 8 x 1 = 8 marks

1. Define centre of mass.
2. Torque magnitude?
3. SI unit of moment of inertia?
4. Rotational kinetic energy?
5. Condition for angular momentum conservation?
6. Rolling relation?
7. Static rigid-body equilibrium?
8. Parallel-axis theorem?

Section B 2-mark questions - 6 x 2 = 12 marks

9. and 3 kg at x=0 and 5 m: x_cm?
10. N perpendicular at .5 m: torque?
11. I=3 kg m^2, omega=4: rotational energy?
12. Radius .4 m wheel at 8 m/s: omega?
13. Torque 12 N m and I=4: alpha?
14. No external force: total momentum?

Section C 3-mark questions - 4 x 3 = 12 marks

15. kg at x=0 and 6 kg at x=4: x_cm?
16. .25 m spanner, 40 N at 30 deg : torque?
17. Solid disc M=4 kg, R=.5 m: I and KE at omega=10?
18. omega rises 2 to 10 rad/s in 4 s: alpha and theta?

Section D 4-mark questions - 2 x 4 = 8 marks

19. kg at x=0 and 3 kg at x=5 m, external force 10 N: x_cm and acceleration?
20. Skater I changes 4 to 1.6 kg m^2 from omega=3: final omega and KE change?